Notebook

1 Introduction

2 Screened longitudinal friction kernel

2.1 Kernel and Gaussian-scale representation

We consider the purely longitudinal matrix-valued kernel

$$K_\kappa(\boldsymbol{r}) = \frac{e^{-\kappa r}}{r} P(\boldsymbol{r}), \qquad P(\boldsymbol{r}) = \frac{\boldsymbol{r}\boldsymbol{r}^\mathsf{T}}{r^2}, \qquad r = |\boldsymbol{r}|, \qquad \kappa > 0. \tag{1}$$

Throughout, $\boldsymbol{k}$ denotes the Fourier wave vector, $k = |\boldsymbol{k}| \geq 0$, and $\xi > 0$ is the Ewald splitting parameter.

Proposition 2.1 (Gaussian-scale representation)**.** *The kernel in Eq. (1) satisfies*

$$K_\kappa(\boldsymbol{r}) = \boldsymbol{r}\boldsymbol{r}^\mathsf{T} \int_0^\infty \rho_\kappa(s) e^{-sr^2}\, ds, \tag{2}$$

with the nonnegative weight

$$\rho_\kappa(s) = \frac{1}{\sqrt{\pi}} \int_0^s t^{-1/2} e^{-\kappa^2/(4t)}\, dt = \frac{2\sqrt{s}}{\sqrt{\pi}} e^{-\kappa^2/(4s)} + \kappa \left[\operatorname{erf}\left(\frac{\kappa}{2\sqrt{s}} \right) - 1 \right]. \tag{3}$$

The proof is given in Appendix A.

2.2 Ewald splitting

We split the Gaussian scales in Eq. (2) at $s = \xi^2$:

$$K_\kappa^L(\boldsymbol{r}) = \boldsymbol{r}\boldsymbol{r}^\mathsf{T} \int_0^{\xi^2} \rho_\kappa(s) e^{-sr^2}\, ds, \tag{4}$$

$$K_\kappa^S(\boldsymbol{r}) = \boldsymbol{r}\boldsymbol{r}^\mathsf{T} \int_{\xi^2}^\infty \rho_\kappa(s) e^{-sr^2}\, ds, \tag{5}$$

o that $K_\kappa = K_\kappa^S + K_\kappa^L$. Since $\boldsymbol{r}\boldsymbol{r}^\mathsf{T} = r^2 P(\boldsymbol{r})$, both pieces remain purely longitudinal,

$$K_\kappa^{S/L}(\boldsymbol{r}) = \theta_{S/L}(r) P(\boldsymbol{r}), \tag{6}$$

where

$$\theta_L(r) = r^2 \int_0^{\xi^2} \rho_\kappa(s) e^{-sr^2}\, ds, \tag{7}$$

$$\theta_S(r) = r^2 \int_{\xi^2}^\infty \rho_\kappa(s) e^{-sr^2}\, ds. \tag{8}$$

Because $\rho_\kappa(s) \geq 0$, both $\theta_L(r) \geq 0$ and $\theta_S(r) \geq 0$, so each real-space piece is pointwise positive semidefinite. Closed-form expressions for θ_L and θ_S are given in Appendix B.

2.3 Localization in real and Fourier space

We use the Fourier convention

$$\widehat{f}(\boldsymbol{k}) = \int_{\mathbb{R}^3} e^{-i\boldsymbol{k}\cdot\boldsymbol{r}} f(\boldsymbol{r})\, d\boldsymbol{r}. \tag{9}$$

Proposition 2.2 (Gaussian localization)**.** *The short-range kernel satisfies*

$$K_\kappa^S(\boldsymbol{r}) \sim \rho_\kappa(\xi^2) e^{-\xi^2 r^2} P(\boldsymbol{r}), \qquad r \to \infty. \tag{10}$$

The long-range kernel has the Fourier transform

$$\widehat{K}_\kappa^L(\boldsymbol{k}) = A_{\kappa,\xi}(k) I + B_{\kappa,\xi}(k) \boldsymbol{k}\boldsymbol{k}^{\mathsf{T}}, \tag{11}$$

where

$$A_{\kappa,\xi}(k) = \frac{\pi^{3/2}}{2} \int_0^{\xi^2} \rho_\kappa(s) s^{-5/2} e^{-k^2/(4s)}\, ds, \tag{12}$$

$$B_{\kappa,\xi}(k) = -\frac{\pi^{3/2}}{4} \int_0^{\xi^2} \rho_\kappa(s) s^{-7/2} e^{-k^2/(4s)}\, ds, \tag{13}$$

and $B_{\kappa,\xi}(k) = A'_{\kappa,\xi}(k)/k$ *for* $k > 0$*. As* $k \to \infty$*,*

$$A_{\kappa,\xi}(k) \sim \frac{2\pi^{3/2}}{\xi k^2} \rho_\kappa(\xi^2) e^{-k^2/(4\xi^2)}, \qquad A_{\kappa,\xi}(k) + k^2 B_{\kappa,\xi}(k) \sim -\frac{\pi^{3/2}}{\xi^3} \rho_\kappa(\xi^2) e^{-k^2/(4\xi^2)}, \tag{14}$$

and consequently

$$\left\| \widehat{K}_\kappa^L(\boldsymbol{k}) \right\|_2 \sim \frac{\pi^{3/2}}{\xi^3} \rho_\kappa(\xi^2) e^{-k^2/(4\xi^2)}, \qquad k \to \infty. \tag{15}$$

The proof is given in Appendix C. The value of $\widehat{K}_\kappa^L$ at $k = 0$ is obtained directly from the integral formulas.

The truncation errors of the two pieces are measured by the continuum tail quantities

$$E_S(r_c) = \int_{|\boldsymbol{r}|>r_c} \|K_\kappa^S(\boldsymbol{r})\|_2\, d\boldsymbol{r}, \qquad E_F(k_c) = \frac{1}{(2\pi)^3} \int_{|\boldsymbol{k}|>k_c} \left\| \widehat{K}_\kappa^L(\boldsymbol{k}) \right\|_2 d\boldsymbol{k}. \tag{16}$$

By Proposition 2.2 (see Appendix C),

$$E_S(r_c) \sim \frac{2\pi}{\xi^2} \rho_\kappa(\xi^2) r_c e^{-\xi^2 r_c^2}, \qquad E_F(k_c) \sim \frac{1}{\sqrt{\pi}\,\xi} \rho_\kappa(\xi^2) k_c e^{-k_c^2/(4\xi^2)}. \tag{17}$$

The quantity E_F is only a continuum asymptotic proxy for the reciprocal tail. A fully discrete periodic PPPM error estimate also depends on the reciprocal lattice, mesh spacing, assignment order, aliasing, and interpolation; these effects are analyzed in Section 3.

2.4 Leading-order balance

Balancing the dominant exponential rates $e^{-\xi^2 r_c^2}$ and $e^{-k_c^2/(4\xi^2)}$ in Eq. (17) gives

$$\xi r_c \approx \frac{k_c}{2\xi}, \qquad \text{or equivalently} \qquad \xi^2 \approx \frac{k_c}{2r_c}. \tag{18}$$

Increasing ξ makes the real-space part more localized but broadens the reciprocal-space contribution, while decreasing ξ does the opposite.

2.5 Pointwise positivity versus Fourier-multiplier signs

The real-space positivity of the split kernels should not be confused with positive semidefiniteness of the Fourier multiplier. By Eq. (11), the eigenvalue of $\widehat{K}_\kappa^L(\boldsymbol{k})$ on vectors orthogonal to $\boldsymbol{k}$ is $\lambda_\perp(k) = A_{\kappa,\xi}(k)$, whereas the eigenvalue in the $\boldsymbol{k}$ direction is

$$\lambda_\parallel(k) = A_{\kappa,\xi}(k) + k^2 B_{\kappa,\xi}(k) = A_{\kappa,\xi}(k) + kA'_{\kappa,\xi}(k). \tag{19}$$

By Eq. (14), $\lambda_\parallel(k) < 0$ for large k, so the Fourier multiplier is generally not positive semidefinite. This does not contradict the pointwise real-space property $K_\kappa^L(\boldsymbol{r}) \succeq 0$. Only the latter enters the graph-Laplacian argument below.

3 Fast periodic graph-Laplacian action

This section concerns the particle graph and its particle–mesh discretization. We use only two properties established in Section 2: the periodic real-space long-range pair kernel $K_\kappa^{L,\mathrm{per}}$ is pointwise positive semidefinite, and its Fourier multiplier has the Gaussian decay of Eq. (15).

3.1 Finite reciprocal truncation and a spectral-gap criterion

For a cubic periodic cell of volume $V = L^3$, let

$$\mathcal{K} = \frac{2\pi}{L}\mathbb{Z}^3. \tag{20}$$

The exact periodic long-range kernel is represented by

$$K_\kappa^{L,\mathrm{per}}(\boldsymbol{r}) = \frac{1}{V}\sum_{\boldsymbol{k}\in\mathcal{K}} \widehat{K}_\kappa^L(\boldsymbol{k}) e^{i\boldsymbol{k}\cdot\boldsymbol{r}}. \tag{21}$$

After a reciprocal cutoff k_c,

$$K_\kappa^{L,k_c}(\boldsymbol{r}) = \frac{1}{V}\sum_{|\boldsymbol{k}|\le k_c} \widehat{K}_\kappa^L(\boldsymbol{k}) e^{i\boldsymbol{k}\cdot\boldsymbol{r}}. \tag{22}$$

Define the omitted tail

$$R_{k_c}(\boldsymbol{r}) = K_\kappa^{L,k_c}(\boldsymbol{r}) - K_\kappa^{L,\mathrm{per}}(\boldsymbol{r}). \tag{23}$$

Then

$$\|R_{k_c}(\boldsymbol{r})\|_2 \le \frac{1}{V}\sum_{|\boldsymbol{k}|>k_c} \|\widehat{K}_\kappa^L(\boldsymbol{k})\|_2. \tag{24}$$

If $\Gamma_L^{k_c}$ is formed from the truncated pair response using the same degree-minus-weight construction, Lemma D.1 in Appendix D gives

$$\|\Gamma_L^{k_c} - \Gamma_L\|_2 \le 2\max_i \sum_{j\neq i} \|R_{k_c}(q_i - q_j)\|_2. \tag{25}$$

A configuration-independent but generally pessimistic consequence is

$$\|\Gamma_L^{k_c} - \Gamma_L\|_2 \le \frac{2(N-1)}{V}\sum_{|\boldsymbol{k}|>k_c} \|\widehat{K}_\kappa^L(\boldsymbol{k})\|_2. \tag{26}$$

The Gaussian Fourier asymptotics imply the expected fixed-system rate

$$\|\Gamma_L^{k_c} - \Gamma_L\|_2 = \mathcal{O}\Big(k_c e^{-k_c^2/(4\xi^2)}\Big), \tag{27}$$

although the worst-case prefactor inherited from Eq. (26) can be very conservative.

Let

$$\mathcal{T} = \mathrm{span}\{\mathbf{1} \otimes e_1, \mathbf{1} \otimes e_2, \mathbf{1} \otimes e_3\}. \tag{28}$$

Assume for the moment that the exact graph has no null vectors beyond translations, and define the restricted spectral gap

$$\lambda_* = \min_{\substack{X \perp \mathcal{T} \\ \|X\|=1}} X^\mathsf{T} \Gamma_L X > 0. \tag{29}$$

Proposition 3.1 (PSD preservation under a small reciprocal perturbation)**.** *If* $\Gamma_L^{k_c}$ *preserves the translation nullspace and*

$$\|\Gamma_L^{k_c} - \Gamma_L\|_2 < \lambda_*, \tag{30}$$

then $\Gamma_L^{k_c} \succeq 0$.

Proof. For $X \perp \mathcal{T}$ with $\|X\| = 1$,

$$X^\mathsf{T} \Gamma_L^{k_c} X = X^\mathsf{T} \Gamma_L X + X^\mathsf{T} (\Gamma_L^{k_c} - \Gamma_L) X \tag{31}$$

$$\geq \lambda_* - \|\Gamma_L^{k_c} - \Gamma_L\|_2 > 0. \tag{32}$$

On $\mathcal{T}$, both operators vanish. □

The condition in Proposition 3.1 is sufficient but not necessary. A truncated pair kernel can fail to be pointwise semidefinite while the corresponding graph Laplacian remains positive semidefinite.

3.2 Symmetric particle–mesh discretization

Let P_h denote spreading from particles to a periodic mesh and let

$$I_h = P_h^\mathsf{T} \tag{33}$$

be the adjoint gathering operator with consistent normalization. Let F_h be the discrete Fourier transform and let $G_h(\boldsymbol{k})$ be an even, real-symmetric tensor influence function. The particle response is

$$W_h = I_h F_h^{-1} G_h F_h P_h. \tag{34}$$

Adjoint transfer and an even symmetric multiplier imply that W_h is symmetric. Rather than using W_h directly as a friction matrix, define

$$(\widetilde{\Gamma}_h^L X)_i = \left(\sum_j (W_h)_{ij} \right) x_i - \sum_j (W_h)_{ij} x_j. \tag{35}$$

The self response cancels algebraically, so the block-row sums vanish exactly and the translation modes are preserved for every mesh and particle configuration. Symmetry and row-zero structure, however, do not by themselves guarantee positive semidefiniteness.

If K_{ij}^h denotes the complete approximate pair response and

$$E_{ij} = K_{ij}^h - K_{ij}, \tag{36}$$

define

$$\epsilon_{\mathrm{op}} = 2 \max_i \sum_{j \neq i} \|E_{ij}\|_2. \tag{37}$$

Lemma D.1 gives

$$\|\widetilde{\Gamma}_h - \Gamma\|_2 \leq \epsilon_{\mathrm{op}}. \tag{38}$$

Therefore, if the exact graph has restricted gap λ_*, defined in Eq. (29), and

$$\epsilon_{\mathrm{op}} < \lambda_*, \tag{39}$$

then the raw particle–mesh graph is already positive semidefinite and no stabilization is required.

If this sufficient certificate is not available, one may retain the translation-preserving fallback stabilization

$$\Gamma_h = \widetilde{\Gamma}_h + \delta_h \Pi, \qquad \Pi = \left(I_N - \frac{1}{N} \mathbf{1}\mathbf{1}^{\mathsf{T}} \right) \otimes I_3, \tag{40}$$

with

$$\delta_h \geq \max \left\{ 0, -\lambda_{\min} \left(\widetilde{\Gamma}_h |_{\mathrm{Range}\, \Pi} \right) \right\}. \tag{41}$$

Proposition 3.2 (Stabilized operator)**.** *The operator Γ_h is symmetric positive semidefinite, satisfies $\Gamma_h t = 0$ for every $t \in \mathcal{T}$, and obeys*

$$\|\Gamma_h - \Gamma\|_2 \leq \epsilon_{\mathrm{op}} + \delta_h. \tag{42}$$

The proof is given in Appendix D. The conservative choice $\delta_h = \epsilon_{\mathrm{op}}$ is always certified; in the tests of Section 6.3 the smallest eigenvalue on $\mathrm{Range}\, \Pi$ moved by at most 41% of $\|\widetilde{\Gamma}_h - \Gamma\|_2$.

3.3 Complete off-grid mesh-response decomposition

The mesh error is most transparent when the aliases are kept explicitly. Let

$$\boldsymbol{k}_n = \boldsymbol{k} + \frac{2\pi}{h} n, \qquad n \in \mathbb{Z}^3, \tag{43}$$

and let $\widehat{W}_h$ be the Fourier transform of the assignment window. For a particle at q, define the deposited amplitude

$$S_h(\boldsymbol{k}, q) = \sum_{n \in \mathbb{Z}^3} \widehat{W}_h(\boldsymbol{k}_n) e^{-i \boldsymbol{k}_n \cdot q}. \tag{44}$$

Separate the principal contribution and the aliases:

$$S_0(\boldsymbol{k}, q) = \widehat{W}_h(\boldsymbol{k}) e^{-i \boldsymbol{k} \cdot q}, \tag{45}$$

$$A_h(\boldsymbol{k}, q) = \sum_{n \neq 0} \widehat{W}_h(\boldsymbol{k}_n) e^{-i \boldsymbol{k}_n \cdot q}. \tag{46}$$

Define the relative alias factor

$$\alpha_h(\boldsymbol{k}, q) = \frac{A_h(\boldsymbol{k}, q)}{\widehat{W}_h(\boldsymbol{k}) e^{-i \boldsymbol{k} \cdot q}}, \tag{47}$$

so that

$$S_h(\boldsymbol{k}, q) = \widehat{W}_h(\boldsymbol{k}) e^{-i \boldsymbol{k} \cdot q} \left[1 + \alpha_h(\boldsymbol{k}, q) \right]. \tag{48}$$

The phase factor

$$e^{-i(2\pi/h) n \cdot q} \tag{49}$$

inside α_h shows explicitly that a finite mesh response depends on particle positions relative to the grid, not only on pair separation.

Define

$$H_h(\boldsymbol{k}) = |\widehat{W}_h(\boldsymbol{k})|^2 G_h(\boldsymbol{k}), \qquad D_h(\boldsymbol{k}) = H_h(\boldsymbol{k}) - \widehat{K}_\kappa^L(\boldsymbol{k}). \tag{50}$$

A direct computation (Appendix E) gives the exact single-mode pair-response error

$$E_{ij}^h(\boldsymbol{k}) = e^{i\boldsymbol{k}\cdot(q_i-q_j)}\left[D_h(\boldsymbol{k}) + H_h(\boldsymbol{k})\left(\alpha_i^* + \alpha_j + \alpha_i^*\alpha_j\right)\right], \tag{51}$$

where $\alpha_i = \alpha_h(\boldsymbol{k}, q_i)$ and similarly for α_j. Equation (51) separates an alias-free influence defect D_h from the complete off-grid alias response.

For ideal deconvolution,

$$G_h(\boldsymbol{k}) = \frac{\widehat{K}_\kappa^L(\boldsymbol{k})}{|\widehat{W}_h(\boldsymbol{k})|^2}, \tag{52}$$

whenever the denominator is numerically safe. Then

$$D_h(\boldsymbol{k}) = 0 \tag{53}$$

in exact arithmetic, and the complete mesh error is produced entirely by the alias cross terms

$$\widehat{K}_\kappa^L(\boldsymbol{k})\left(\alpha_i^* + \alpha_j + \alpha_i^*\alpha_j\right). \tag{54}$$

The linear terms are first order in the alias amplitude, while the quadratic term is second order. This distinction is important when defining numerical diagnostics: a diagnostic retaining only matched quadratic aliases can scale like h^{2p} and miss the dominant h^p contribution.

3.4 B-spline assignment and the h^p mesh law

For a cardinal B-spline assignment of order p,

$$\widehat{W}_h(\boldsymbol{k}) = \prod_{\ell=1}^{3} \operatorname{sinc}^p\left(\frac{k_\ell h}{2}\right), \qquad \operatorname{sinc}(x) = \frac{\sin x}{x}. \tag{55}$$

For $p \geq 2$, the relative alias amplitude satisfies

$$\alpha_h(\boldsymbol{k}, q) = \mathcal{O}((h|\boldsymbol{k}|)^p). \tag{56}$$

Because the screened long-range Fourier multiplier has Gaussian decay, all moments

$$M_p(\kappa, \xi) = \frac{1}{(2\pi)^3}\int_{\mathbb{R}^3} k^p \|\widehat{K}_\kappa^L(\boldsymbol{k})\|_2 \, d\boldsymbol{k} \tag{57}$$

are finite. Therefore the complete off-grid mesh pair response obeys the asymptotic law

$$\epsilon_{\text{mesh}} = \mathcal{O}(h^p) \tag{58}$$

for fixed physical cutoff and sufficiently fine mesh; the proof is given in Appendix E.

The phase dependence in α_h implies oscillations in a single fixed configuration as h varies. Random global grid offsets remove the mean of the linear phase terms but not their RMS magnitude. Consequently, RMS phase-averaged errors give a cleaner estimate of the asymptotic h^p behavior.

It is useful to write the measured asymptotic law in dimensionless form as

$$\epsilon_{\text{mesh}} \approx C_M(\kappa, \xi, p, k_c, \text{configuration})(\xi h)^p. \tag{59}$$

The coefficient in Eq. (59) is not universal; it depends on the configuration and on the reciprocal cutoff. Parameter selection should therefore use a conservative envelope $C_M^{\max}(p)$ rather than a single universal constant.

4 Discrete FDT thermostat and time integration

4.1 A finite-time thermostat map

The matrix-free Ewald–PPPM formulation of Section 3 provides a fast action of the discrete friction operator,

$$\boldsymbol{x} \mapsto \Gamma_h \boldsymbol{x}, \tag{60}$$

whose cost at fixed splitting parameters and fixed density, with a mesh of $M^3 \propto N$ points, is

$$T_{\Gamma_h}(N) = \mathcal{O}(N \log N). \tag{61}$$

The remaining difficulty is the correlated thermal noise. For the exponential finite-time thermostat, the damping and noise factors are

$$R_{\Delta t}(\Gamma_h) = \exp\left(-\frac{\Delta t}{m}\Gamma_h\right), \tag{62}$$

and

$$S_{\Delta t}(\Gamma_h) = \left[\beta^{-1} m \left(I - R_{\Delta t}(\Gamma_h)^2\right)\right]^{1/2}. \tag{63}$$

The pair $(R_{\Delta t}, S_{\Delta t})$ reproduces the exact transition law of the frozen-configuration Ornstein–Uhlenbeck flow, conserves total momentum pathwise, and leaves the conditional Maxwell law invariant; see Appendix F. The random increment is therefore

$$\eta = f_{\Delta t}(\Gamma_h)\, \xi, \tag{64}$$

with

$$f_{\Delta t}(\lambda) = \sqrt{\beta^{-1} m \left(1 - e^{-2\Delta t \lambda / m}\right)}. \tag{65}$$

We evaluate this matrix-function action by the Lanczos method. If the required Krylov rank is $r(N)$, the noise cost is

$$T_{\mathrm{noise}}(N) = \mathcal{O}(r(N) N \log N)\,. \tag{66}$$

Consequently, an $\mathcal{O}(N \log N)$ cost per step requires that the rank remain effectively constant under a fixed accuracy budget.

4.2 Lanczos evaluation of the correlated noise

We first consider direct-start Lanczos with

$$z = \Pi \xi, \qquad \Pi = \left(I - \frac{1}{N}\mathbf{1}\mathbf{1}^{\mathsf{T}}\right) \otimes I_3, \tag{67}$$

and

$$q_1 = \frac{z}{\|z\|}, \qquad \eta_r = \|z\| Q_r f_{\Delta t}(T_r) e_1. \tag{68}$$

The projection Π is applied after every Γ_h action and after the final reconstruction.

4.3 Range-start Lanczos

A Landau-style range start is a natural alternative. Defining

$$b = \Gamma_h z, \tag{69}$$

and

$$g(\lambda) = \frac{f_{\Delta t}(\lambda)}{\lambda}, \tag{70}$$

one may write, on the positive spectral range,

$$f_{\Delta t}(\Gamma_h) z = g(\Gamma_h) b. \tag{71}$$

Although $b = \Gamma_h z$ strongly suppresses the smallest spectral modes in the starting vector, the transformed scalar function behaves as

$$g(\lambda) \sim C\lambda^{-1/2}, \qquad \lambda \to 0^+, \tag{72}$$

because

$$f_{\Delta t}(\lambda) \sim C\sqrt{\lambda}, \tag{73}$$

with $C = (2\beta^{-1}\Delta t)^{1/2}$ (Appendix F). The increased singularity of g can offset the spectral reweighting.

4.4 Full symmetric splitting

The PPPM–Lanczos thermostat is inserted into the symmetric Langevin splitting

$$\Psi_{\Delta t} = B_{\Delta t/2} A_{\Delta t/2} O_{\Delta t} A_{\Delta t/2} B_{\Delta t/2}. \tag{74}$$

For the ideal gas, $U = 0$ and the conservative B steps vanish. One step reduces to

$$q^{(1)} = q^n + \frac{\Delta t}{2m} p^n, \tag{75}$$

$$p^{n+1} = e^{-\Delta t \Gamma_h(q^{(1)})/m} p^n + f_{\Delta t}(\Gamma_h(q^{(1)}))\xi^n, \tag{76}$$

and

$$q^{n+1} = q^{(1)} + \frac{\Delta t}{2m} p^{n+1}, \tag{77}$$

with periodic wrapping of the positions.

Both the deterministic damping action and the stochastic noise action are evaluated matrix-free. The center-of-mass momentum component is copied exactly through the damping step, while the fluctuating component and every noise sample are projected onto Range(Π).

5 Simulation error and complexity

5.1 Operator error budget

For the present ideal-deconvolution implementation, it is more natural to combine aliasing and off-grid transfer into one complete mesh-response term:

$$\epsilon_{\text{op}} \le \epsilon_S + \epsilon_F + \epsilon_{\text{mesh}} + \epsilon_{\text{quad}}. \tag{78}$$

For the screened longitudinal kernel considered here the Fourier multiplier is available analytically, so ϵ_{quad} can be made negligible. With the balanced variable

$$s = \xi r_c = \frac{k_c}{2\xi}, \tag{79}$$

the real- and reciprocal-space Ewald tails have the common Gaussian rate

$$\epsilon_S + \epsilon_F \sim C_E e^{-s^2} \tag{80}$$

up to algebraic prefactors, whereas for fixed B-spline order

$$\epsilon_{\text{mesh}} \approx C_M (\xi h)^p. \tag{81}$$

If the mesh is parameterized by

$$k_c = \eta_N \frac{\pi}{h}, \qquad 0 < \eta_N < 1, \tag{82}$$

then

$$\xi h = \frac{\eta_N \pi}{2s}, \tag{83}$$

and the schematic total error becomes

$$\epsilon_{\text{op}} \lesssim C_E e^{-s^2} + C_M \left(\frac{\eta_N \pi}{2s}\right)^p. \tag{84}$$

Thus, in Eq. (84), the Ewald tail is Gaussian in s, while the mesh error is only algebraic in s at fixed assignment order. At sufficiently high target accuracy the mesh term must therefore dominate unless p is increased or h is reduced more aggressively.

5.2 Practical parameter selection

A convenient initial Ewald choice is

$$s = \sqrt{\log\left(\frac{2}{\text{tol}}\right)}. \tag{85}$$

The balanced cutoffs are then

$$r_c = \frac{s}{\xi}, \qquad k_c = 2\xi s. \tag{86}$$

The factor ξ should be treated as a computational work-balancing parameter rather than an accuracy parameter. For a chosen B-spline order p, a practical mesh requirement is

$$C_M^{\max}(p)(\xi h)^p \le \frac{\text{tol}}{2}, \tag{87}$$

which gives

$$h \le \frac{1}{\xi}\left(\frac{\text{tol}}{2C_M^{\max}(p)}\right)^{1/p}. \tag{88}$$

This should be combined with a Nyquist safety constraint such as

$$\frac{k_c h}{\pi} \le 0.8. \tag{89}$$

For a box of side length L, one can then choose

$$M \geq \frac{L}{h} \tag{90}$$

and round upward to an FFT-friendly size. This last rounding decreases the actual mesh spacing and therefore does not weaken the accuracy condition.

Once a complete pair-response estimate is available, accuracy is certified through

$$\|\widetilde{\Gamma}_h - \Gamma\|_2 \leq \epsilon_{\mathrm{op}}. \tag{91}$$

If a reliable restricted gap λ_* is also available, the same error estimate provides the optional PSD certificate

$$\epsilon_{\mathrm{op}} < \lambda_* \quad \Longrightarrow \quad \widetilde{\Gamma}_h \succeq 0. \tag{92}$$

Failure of this sufficient inequality does not imply indefiniteness; it only means that the perturbation certificate is inconclusive.

5.3 Work estimate

With fixed noise and damping ranks r_{noise} and r_{damp}, the dominant cost per thermostat step can be summarized as

$$T_{\mathrm{step}} = (r_{\mathrm{noise}} + r_{\mathrm{damp}})\, T_{\Gamma_h} + T_{\mathrm{setup}} + T_{\mathrm{reorth}} + T_{\mathrm{small\ matrix}} + T_{\mathrm{vector}}. \tag{93}$$

If the ranks remain bounded under a fixed accuracy budget and $T_{\Gamma_h} = \mathcal{O}(N \log N)$, the cost per step is $\mathcal{O}(N \log N)$; if the ranks must grow with N, the cost grows with $r_{\mathrm{noise}} + r_{\mathrm{damp}}$. Whether the ranks remain bounded is not established here.

6 Numerical experiment: the particle–mesh operator

This section tests whether the operator applied by the method, $X \mapsto \widetilde{\Gamma}_h X$, reproduces the periodic friction operator Γ of the kernel of Eq. (1) with the accuracy expected from (ξ, r_c, k_c, h, p). The kernel is used as defined, with $\kappa = 1$; the friction is fully periodic and the $k = 0$ Fourier mode is retained in both the reference and the particle–mesh operator. The configurations are static, so no time-discretization or Lanczos error enters. Each error source of Eq. (78) is first measured on its own (Section 6.2); the complete production action is then compared with the reference (Section 6.3). All numbers come from the result files of the project repository (`ewald_longitudinal/exp1_results/`), computed after the correction of a wave-number rounding in the coefficient routine described in Appendix G.

6.1 Configurations, reference and parameter sets

Every error is measured against a dense $3N \times 3N$ reference and its spectrum, which restricts the test to $N \leq 256$. Table 1 lists the 44 configurations. Cases A and B are the two boxes on which the parameter sets below were accepted. The main set has $N = 128$ at the density $n_A = 30/216$ of case A, with three families and three seeds each: independent uniform positions, perturbed lattices, and clusters; one configuration per family at $N = 256$ forms a check set. Two box-length sets use uniform positions with three seeds per box: at fixed $N = 64$, where the density changes with L, and at fixed density n_A, where N changes with L. The generators are described in Appendix G.

set	generation	N	L	N/L^3	$d_{\min}$	seeds
A	uniform	30	6	0.139	0.395	1
B	uniform	40	10	0.040	0.589	1
uniform	uniform	128	9.73	0.139	0.142–0.411	3
lattice	perturbed bcc	128	9.73	0.139	1.208–1.366	3
clustered	8 clusters of 16	128	9.73	0.139	0.251–0.255	3
check	one per family	256	12.26	0.139	0.148–1.011	1
fixed N	uniform	64	4–16 (6 values)	0.016–1.00	0.090–0.723	3
fixed density	uniform	30–240	6–12 (4 values)	0.139	0.055–0.924	3

Table 1: Static configurations, $\kappa = 1$. L is the edge of the periodic cube and $d_{\min}$ the smallest periodic pair distance (range over seeds). The lattices are bcc with 4^3 cells at $N = 128$ and fcc with 4^3 cells at $N = 256$. Source: `configs.csv`.

Reference. The reference for Γ is assembled from the unsplit kernel by a direct periodic lattice sum over the images $|\boldsymbol{r}_{ij} + \boldsymbol{m}L| \le R$, $\boldsymbol{m} \in \mathbb{Z}^3$, with no minimum-image approximation and no splitting; nothing is removed from the sum, so the $k = 0$ mode is included. Increasing R from 15 to 45, the largest change relative to $R = 45$, in $\|\cdot\|_2/\|\Gamma\|_2$ over all 44 configurations, is 2.5×10^{-12} at $R = 30$, 2.0×10^{-14} at $R = 35$ and 4.3×10^{-16} at $R = 40$; $R = 45$ is used. This ladder is numerical evidence of convergence, not a bound on the truncation error. An independent construction (the split identity below) reproduces the reference to 1.1×10^{-15}. The reference is symmetric to 1.0×10^{-16} and annihilates $\mathcal{T}$ to 1.9×10^{-16}, both relative; $\|\Gamma\|_2$ ranges from 0.72 to 35 over the configurations.

Parameter sets. Three sets were tested (Table 2). They are the cheapest sets at $\xi = 0.7$ that met the relative operator tolerances 10^{-5}, 10^{-7} and 10^{-9} on cases A and B, and are called T5, T7 and T9. Their tolerances were calibrated on these two boxes only; whether they hold elsewhere is what Section 6.3 tests.

set	s	$r_c = s/\xi$	$k_c = 2\xi s$	η_N, p	M (A / B / $N = 128$ / $N = 256$)
T5	3.494	4.991	4.891	0.8, 6	12 / 20 / 20 / 24
T7	4.100	5.857	5.740	0.7, 7	16 / 27 / 27 / 36
T9	4.628	6.611	6.479	0.6, 8	24 / 36 / 36 / 45

Table 2: Parameter sets, $\xi = 0.7$. M is the smallest $2, 3, 5$-smooth mesh size with $h \le \eta_N \pi/k_c$, so the realised $k_c h/\pi$ is at most η_N and depends on L. All sets retain $\boldsymbol{k} = 0$ and exclude no influence mode. Source: `action.csv`.

6.2 Error sources measured separately

Split identity. With θ_S in closed form and θ_L from its own s-integral, the identity $\theta_S + \theta_L = e^{-\kappa r}/r$ holds to a relative error 2.5×10^{-15} for $\xi \in \{0.5, 0.7, 1, 1.4, 2\}$, on a radial grid $10^{-3} \le r \le 20$ and on every pair image with $r \le 12$. At operator level, the real-space lattice sum of θ_S plus the mesh-free lattice Fourier sum of $\widehat{K}_\kappa^L$ (including $\boldsymbol{k} = 0$, $k_c = 13\xi$) reproduces the reference to 1.4×10^{-16}–1.1×10^{-15} of $\|\Gamma\|_2$ on cases A, B and the main set for $\xi \in \{0.5, 0.7, 1, 1.4\}$. The largest numerical discrepancy found in these checks and in the floor analysis of Appendix G is 1.1×10^{-15}; in all fits below, only errors of at least 100 times this value, 1.1×10^{-13}, are used (the fit floor).

Truncation. Figure 1 shows the real-space error, the operator-norm distance between the real-space lattice sum of θ_S truncated at r_c and the untruncated one, and the Fourier error $\|\Gamma_L^{k_c} - \Gamma_L\|_2$, each with no other approximation present and normalized by $\|\Gamma\|_2$. Fitting $\log E = a - cx^2$ with $x = \xi r_c$ or $x = k_c/(2\xi)$ on the points above the fit floor gives, at $\xi = 0.7$, median rates $c = 0.99$ (real space, range 0.98–1.02 over cases A, B and the main set) and $c = 0.97$ (Fourier, 0.96–1.00); at $\xi = 1$ the medians are 0.99 and 0.98. Both truncations therefore decay at the Gaussian rate e^{-x^2} of Eq. (17) over the fitted window. Dividing by the continuum prefactor x overcorrects (medians 1.03–1.05), so the algebraic prefactor of the operator-norm error is not that of the continuum tail, and c is not interpreted as an asymptotic constant.

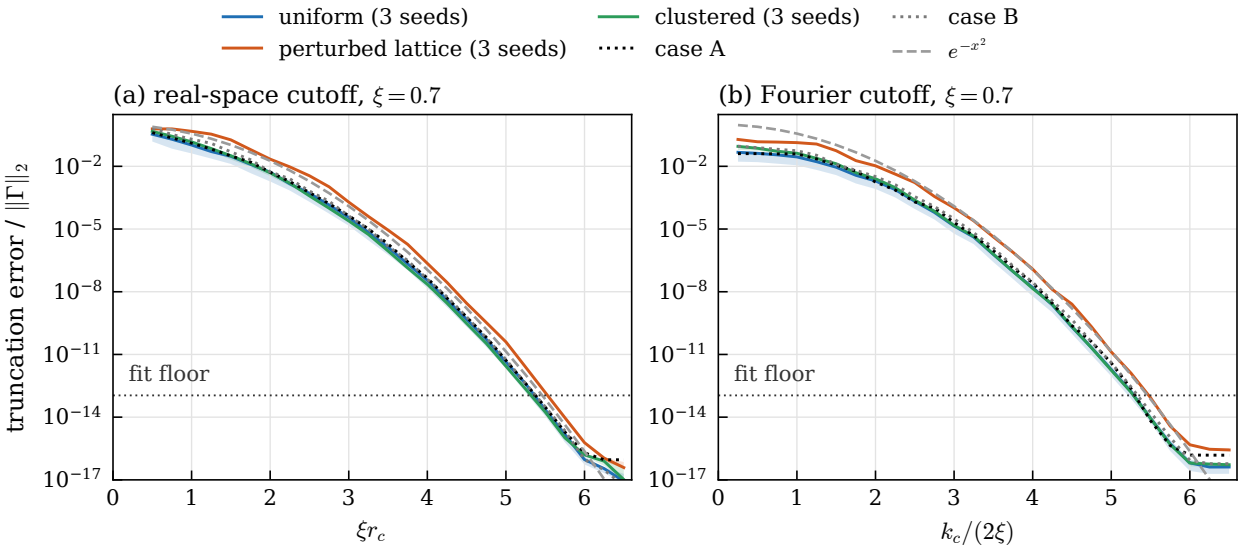


Figure 1: Truncation errors at $\xi = 0.7$, each with no other approximation. (a) Real-space cutoff; (b) Fourier cutoff $\|\Gamma_L^{k_c} - \Gamma_L\|_2$. Both are normalized by $\|\Gamma\|_2$. Solid curves: median over three seeds of each main-set family; bands: range over seeds; dotted: cases A and B; dashed: e^{-x^2}; horizontal dotted line: fit floor. Below the fit floor the Fourier error levels off at the numerical floor of Appendix G.

Mesh. At the T7 cutoff $k_c = 5.74$ the long-range particle–mesh operator $\widetilde{\Gamma}_h^L$ of Eq. (35) is compared with the mesh-free sum $\Gamma_L^{k_c}$ over the same modes, so the difference is the mesh error alone. The scan covers $p = 3, \ldots, 8$, nine meshes from $k_c h/\pi \approx 0.91$–0.99 down to 0.23–0.25, and eight grid shifts per mesh (the unshifted grid and seven random global offsets), on cases A, B and the main set (Fig. 2a). On the four finest meshes above the fit floor ($k_c h/\pi \approx 0.23$–0.50) the median fitted slopes in h are 3.09, 4.07, 5.24, 6.19, 7.46, 8.50 for $p = 3, \ldots, 8$ (Fig. 2b); a repetition at the larger cutoff $s = 5.5$ gives 4.10, 6.25, 8.51 for $p = 4, 6, 8$. These slopes are consistent with the h^p law of Section 3. On coarser meshes the fitted exponent is 1.10–1.27 p, consistent with the factor q^p of the leading alias, whose local exponent approaches p only as $h \to 0$. A fit over finitely many meshes does not establish the asymptotic order, and no asymptotic constant is reported. Over the eight grid shifts the largest error is at most 1.30 times the RMS.

In the exact off-grid decomposition of Eq. (51), evaluated on the unshifted grid for $p = 4, 7$ and two meshes on cases A, B and one seed of each main family, the linear terms $\alpha_i^* + \alpha_j$ carry 0.996–1.003 of the mesh error in the spectral norm, the quadratic term $\alpha_i^* \alpha_j$ at most 0.9%, and

D_h at most 4.3×10^{-9} of it, as expected for ideal deconvolution. A diagnostic that kept only the quadratic term would miss the mesh error almost entirely.

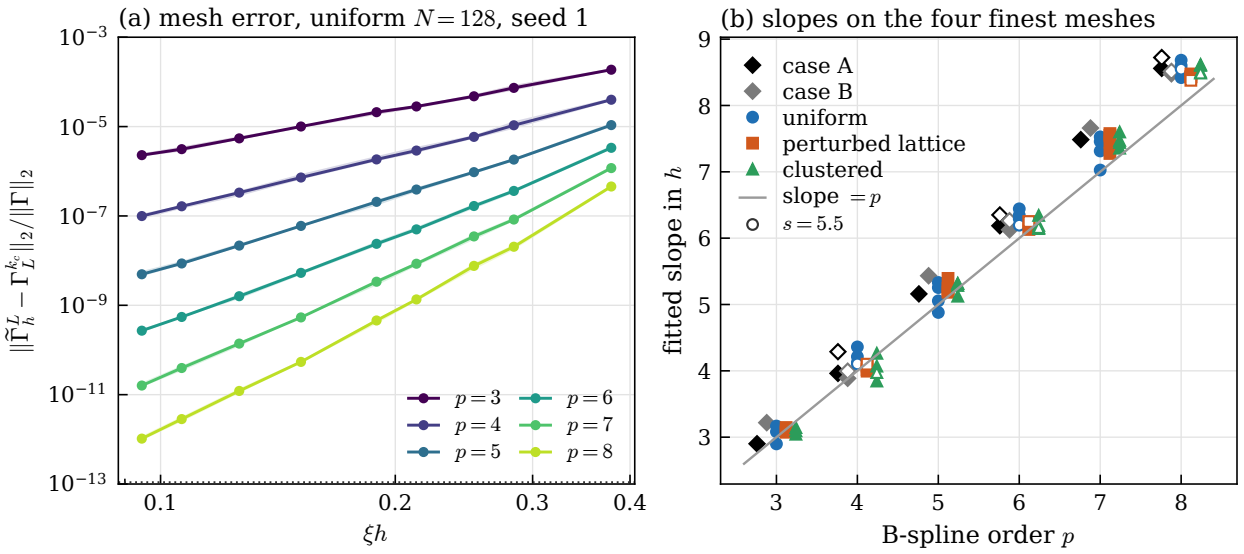


Figure 2: Mesh error at $k_c = 5.74$ ($s = 4.1$, $\xi = 0.7$). (a) $\|\widetilde{\Gamma}_h^L - \Gamma_L^{k_c}\|_2/\|\Gamma\|_2$ versus ξh for the uniform configuration $N = 128$, seed 1: RMS over eight grid shifts (points) and range over shifts (bands); dotted: fit floor. (b) Fitted slopes in h on the four finest meshes above the fit floor, one point per configuration and p (filled: $s = 4.1$; open: $s = 5.5$, five configurations); line: slope p.

6.3 The complete production action

The production code path (real-space pair loop, B-spline spreading, FFT, multiplication by the influence function on $|\boldsymbol{k}| \le k_c$, inverse FFT, adjoint gathering and the degree correction) is applied to the $3N$ unit vectors, giving $\widetilde{\Gamma}_h$ column by column, and to probe vectors. The columns agree with an independent mode-space assembly to 1.1×10^{-15}, and the separately measured terms ϵ_S, ϵ_F, ϵ_{mesh} of Eq. (78) add up to 1.2–2.0 times the measured error. Three quantities are kept apart: the relative operator error $\|\widetilde{\Gamma}_h - \Gamma\|_2/\|\Gamma\|_2$, on which the sets were accepted; the absolute operator error $\|\widetilde{\Gamma}_h - \Gamma\|_2$; and the relative action error $\|(\widetilde{\Gamma}_h - \Gamma)X\|/\|\Gamma X\|$ for a given vector X.

Relative operator error. Table 3 and Fig. 3a compare the relative operator error with the tolerance for which each set was accepted. T5 and T9 were tested on the 14 configurations of cases A, B, the main and the check sets, T7 also on the 30 box-length configurations. Cases A and B meet their tolerances, as selected. The other configurations do not all do so: T5 meets 10^{-5} on 10 of 14 configurations, with largest relative operator error 1.2875×10^{-5}; T7 meets 10^{-7} on 39 of 44, with largest 2.1056×10^{-7}; T9 meets 10^{-9} on 10 of 14, with largest 2.5451×10^{-9}. The exceedances are at most a factor 2.55. They occur on the perturbed lattices for all three sets, and at T7 also on one uniform box of the fixed-density set ($N = 30$, $L = 6$, seed 3; 1.40×10^{-7}), which has the size, density and generator of case A and differs from it only in the seed. The nominal tolerances are therefore not guaranteed beyond the two configurations on which they were calibrated. For the production set T7 the measured statement is: on the 44 static configurations tested, the relative operator error does not exceed 2.2×10^{-7}.

set	tolerance	meets / tested	largest rel. error	absolute error	exceeding
T5	10^{-5}	10 / 14	1.2875×10^{-5}	7.7×10^{-6}–3.3×10^{-5}	lattices
T7	10^{-7}	39 / 44	2.1056×10^{-7}	5.7×10^{-8}–5.5×10^{-7}	lattices, one uniform box
T9	10^{-9}	10 / 14	2.5451×10^{-9}	1.3×10^{-9}–7.1×10^{-9}	lattices

Table 3: Relative operator error $\|\widetilde{\Gamma}_h - \Gamma\|_2 / \|\Gamma\|_2$ against the tolerance for which the set was accepted on cases A and B. "Lattices": the three perturbed lattices at $N = 128$ and the one at $N = 256$. Source: `parameter_set_status.json`.

Both the numerator and the denominator of the relative error depend on the configuration (Fig. 3b). At T7 the absolute error varies by about a factor ten over the 44 configurations, from 5.7×10^{-8} to 5.5×10^{-7}, and is largest for the densest fixed-N boxes ($L = 4$) and the clustered configurations. $\|\Gamma\|_2$ is strongly influenced by close pairs, but not set by the nearest pair alone: with $\theta(r) = e^{-\kappa r}/r$, the ratio $\|\Gamma\|_2/\theta(d_{\min})$ is 2.0–4.1, whereas an isolated pair gives 2. Every configuration that exceeds its tolerance has a small norm (0.72–0.94 for the lattices, 1.50 for the uniform box) and a moderate absolute error. A small norm is not sufficient, however: the fixed-N box with $L = 16$, seed 1, has $\|\Gamma\|_2 = 1.34$ and meets 10^{-7} because its absolute error is smaller.

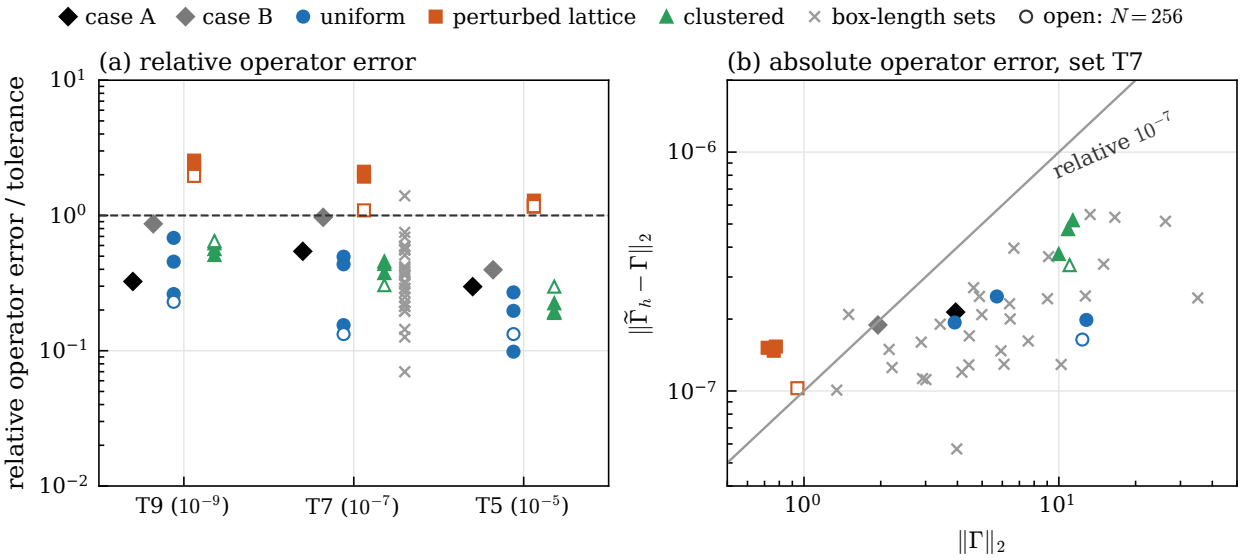


Figure 3: The production operator against the reference, 44 static configurations, one point per configuration. (a) Relative operator error divided by the tolerance for which the set was accepted on cases A and B; dashed: tolerance. Box-length sets were tested with T7 only. (b) Absolute operator error against $\|\Gamma\|_2$ at T7; the line marks a relative error 10^{-7}.

Error budget. For cases A, B and the main set at T7 the mesh term is the largest ($\epsilon_{\text{mesh}} \leq 1.5 \times 10^{-7}$ relative), followed by the real-space and Fourier truncations ($\leq 1.05 \times 10^{-7}$ and $\leq 7 \times 10^{-8}$, largest for the lattices). The real-space term is the largest for the $N = 256$ lattice and for five boxes at the smallest realised $k_c h/\pi = 0.61$.

Action on probe vectors. The probes are sixteen mean-free Gaussian vectors, the modes of the lowest nonzero wave number $2\pi/L$ along each axis (longitudinal and transverse, projected onto $\operatorname{Range}\Pi$), and the translations (Fig. 4a). A relative action error can exceed the relative operator error by the factor $\|\Gamma\|_2\|X\|/\|\Gamma X\|$, which is large for slowly varying vectors: $\|\Gamma X\|/(\|\Gamma\|_2\|X\|)$ is as small as 2.5×10^{-2} (longitudinal) and 1.1×10^{-2} (transverse) on cases A, B and the main set, where these probes exceed their configuration's relative operator error by factors up to 14 and 18. Meeting the operator tolerance therefore does not imply meeting it for every action. At T7, all 39 configurations with relative operator error at most 10^{-7} have a transverse probe with relative action error above 10^{-7}; 32 have such a longitudinal and 21 such a Gaussian probe. Among these 39, the largest relative action errors are 2.2, 3.7 and 4.5 times 10^{-7} (Gaussian, longitudinal, transverse). At T5 and T9, 8 and 10 of the 10 configurations within the operator tolerance have a transverse probe above it, by up to factors 2.5 and 4.9. The production operator annihilates the translations exactly in floating point, because the degree correction cancels the constant field term by term; the reference residual is at most 1.9×10^{-16}.

Symmetry and spectrum. $\widetilde{\Gamma}_h$ is symmetric to 1.0×10^{-16} (relative Frobenius norm). The smallest eigenvalue of Γ on $\operatorname{Range}\Pi$ ranges from 7.9×10^{-4} to 1.92 over the configurations, and $\lambda_{\min}(\widetilde{\Gamma}_h|_{\operatorname{Range}\Pi})$ differs from it by at most 4.2×10^{-5} (T5), 1.6×10^{-6} (T7) and 2.6×10^{-8} (T9), relative. In every one of the 72 tested pairs of configuration and set, $\|\widetilde{\Gamma}_h - \Gamma\|_2 < \lambda_{\min}(\Gamma|_{\operatorname{Range}\Pi})$, so the raw particle–mesh graph is positive semidefinite on these configurations without stabilization; the observed shift of the smallest eigenvalue is a median 13% and at most 41% of $\|\widetilde{\Gamma}_h - \Gamma\|_2$. These are observations on finitely many configurations, not a general positivity result.

Box length and density. At T7 the two box-length sets separate different effects (Fig. 4b; medians over three seeds). At fixed $N = 64$ the density and κL change together; the absolute error falls from 5.3×10^{-7} ($L = 4$, density 1) to 1.0×10^{-7} ($L = 16$, density 0.016), with all three budget terms decreasing, though not monotonically. At fixed density N grows with L, and the absolute error is 2.1×10^{-7}, 1.1×10^{-7}, 2.5×10^{-7} and 2.5×10^{-7} for $L = 6$, 8, 10, 12, a change of at most a factor 2.2. In both sets the dip at $L = 8$ coincides with the smaller realised $k_c h/\pi = 0.61$ caused by rounding M to a smooth size; for $p = 7$ the h^p trend predicts a factor $(0.61/0.68)^7 \approx 0.45$ in the mesh term. This was not rerun at matched $k_c h/\pi$, and the set does not separate the effect of L from that of N and the particle arrangement.

6.4 Summary

On static configurations with $\kappa = 1$ and $N \leq 256$ the experiment establishes the following. The dense reference converges numerically under image refinement and is reproduced by the split construction to 1.1×10^{-15}. Both truncations decay at the Gaussian rate over the fitted window, and the mesh error has slopes consistent with h^p on the finest meshes, carried almost entirely by the linear alias terms. The production action reproduces the separately measured budget, is symmetric, annihilates the translations exactly and remained positive semidefinite in every tested case. The parameter sets calibrated on cases A and B do not meet their nominal relative tolerance on every configuration: 4 of 14 (T5), 5 of 44 (T7) and 4 of 14 (T9) exceed it, by up to a factor 2.55. Even where the operator tolerance is met, relative action errors of slowly varying vectors exceed it by up to a factor 4.9. Parameters therefore have to be checked on configurations representative of their use, and the action on slow modes has to be checked separately from the operator norm.

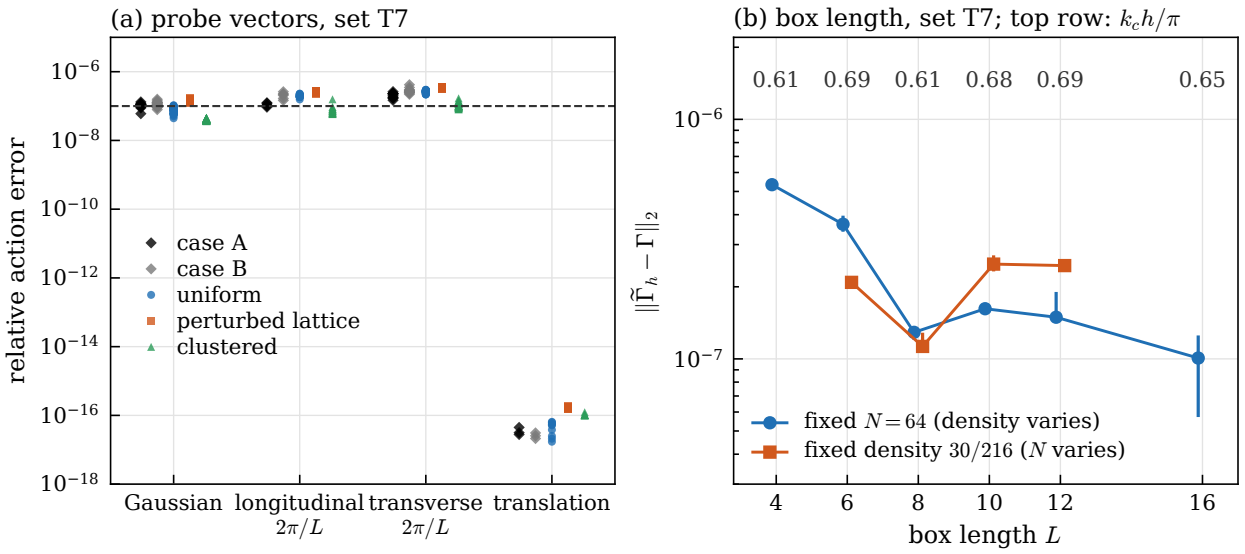


Figure 4: (a) Relative action errors $\|(\widetilde{\Gamma}_h - \Gamma)X\|/\|\Gamma X\|$ of probe vectors at T7 through the production action, cases A, B and the main set; dashed: 10^{-7}. For the translations $t \in \mathcal{T}$, where $\Gamma t = 0$, the quantity shown is $\|(\widetilde{\Gamma}_h - \Gamma)t\|/(\|\Gamma\|_2\|t\|)$, which equals the reference residual because $\widetilde{\Gamma}_h t = 0$ exactly. (b) Absolute operator error at T7 against box length; points: median over three seeds, bars: range; top row: realised $k_c h/\pi$, which is the same in both sets at equal L.

7 Conclusion

Sections 2–5 address the consequences of the screened longitudinal Ewald long-range kernel for the particle graph and its PPPM discretization. The analysis rests on two properties: the real-space long-range pair kernel is pointwise positive semidefinite, and its Fourier multiplier has Gaussian decay $\mathcal{O}(e^{-k^2/(4\xi^2)})$. The Fourier multiplier itself need not be positive semidefinite; positivity of the exact friction follows instead from the pointwise semidefinite pair kernel and the graph degree-minus-weight structure.

Finite reciprocal and particle–mesh approximations preserve symmetry and the translation nullspace when adjoint spreading/gathering and a common degree-minus-weight construction are used. Their operator error is controlled by the complete pair-response discrepancy. For ideal particle–mesh deconvolution, the leading mesh error is generated by the linear off-grid alias cross terms and satisfies

$$\epsilon_{\text{mesh}} = \mathcal{O}(h^p) \tag{94}$$

for a B-spline assignment of order p. The resulting practical error model is

$$\epsilon_{\text{op}} \lesssim C_E e^{-s^2} + C_M(\xi h)^p, \qquad s = \xi r_c = \frac{k_c}{2\xi}. \tag{95}$$

A useful initial choice is

$$s = \sqrt{\log(2/\text{tol})}, \qquad r_c = \frac{s}{\xi}, \qquad k_c = 2\xi s, \tag{96}$$

followed by a mesh chosen from the measured envelope of C_M and a Nyquist safety condition. Accuracy is certified by a complete pair-response operator bound, while

$$\epsilon_{\text{op}} < \lambda_* \tag{97}$$

provides a sufficient, although conservative, certificate that the raw discrete graph remains positive semidefinite.

The operator experiment of Section 6, on static configurations with $\kappa = 1$ and $N \le 256$, supports the components of this error model: both truncations decay at the Gaussian rate, the mesh error has slopes consistent with h^p on the finest meshes and is carried by the linear alias terms, and the production action reproduces the separately measured budget, preserves symmetry and the translation null space, and remained positive semidefinite in every tested case. It also shows the limits of a calibration on few configurations. Parameter sets accepted for the relative tolerances 10^{-5}, 10^{-7} and 10^{-9} on two boxes met them on 10 of 14, 39 of 44 and 10 of 14 configurations, respectively, with largest relative operator errors 1.2875×10^{-5}, 2.1056×10^{-7} and 2.5451×10^{-9}; for the set T7 the measured statement is a relative operator error of at most 2.2×10^{-7} on the 44 configurations tested. Relative action errors on slowly varying vectors can exceed the relative operator error severalfold. These are finite-sample observations; positivity and the observed slopes are not general results.

A Gaussian-scale representation

This appendix proves Proposition 2.1.

Gaussian representations of the two factors. We factor the scalar part of the kernel as

$$\frac{e^{-\kappa r}}{r^3} = \frac{e^{-\kappa r}}{r} \cdot \frac{1}{r^2} \tag{98}$$

and represent each factor as a superposition of Gaussians in r. For the Yukawa factor we use the standard representation

$$\frac{e^{-\kappa r}}{r} = \frac{1}{\sqrt{\pi}} \int_0^\infty t^{-1/2} e^{-\kappa^2/(4t)} e^{-tr^2}\, dt, \tag{99}$$

which is the integral $\int_0^\infty t^{-1/2} e^{-at-b/t}\, dt = \sqrt{\pi/a}\, e^{-2\sqrt{ab}}$, $a, b > 0$, evaluated at $a = r^2$ and $b = \kappa^2/4$. For the second factor,

$$\frac{1}{r^2} = \int_0^\infty e^{-ur^2}\, du. \tag{100}$$

Thus the Gaussian-scale weights of the two factors are $\pi^{-1/2} t^{-1/2} e^{-\kappa^2/(4t)}$ and the constant 1.

Convolution of the weights. Multiplying the two representations, the Gaussians combine as $e^{-tr^2} e^{-ur^2} = e^{-(t+u)r^2}$, so the product is again a superposition of Gaussians, with scale $s = t + u$. Its weight is the Laplace convolution of the two weights. Explicitly, by Tonelli's theorem, since all integrands are nonnegative,

$$\frac{e^{-\kappa r}}{r^3} = \frac{1}{\sqrt{\pi}} \int_0^\infty \int_0^\infty t^{-1/2} e^{-\kappa^2/(4t)} e^{-(t+u)r^2}\, du\, dt \tag{101}$$

$$= \frac{1}{\sqrt{\pi}} \int_0^\infty \int_t^\infty t^{-1/2} e^{-\kappa^2/(4t)} e^{-sr^2}\, ds\, dt \tag{102}$$

$$= \int_0^\infty \left[\frac{1}{\sqrt{\pi}} \int_0^s t^{-1/2} e^{-\kappa^2/(4t)} \cdot 1\, dt \right] e^{-sr^2}\, ds \tag{103}$$

$$= \int_0^\infty \rho_\kappa(s) e^{-sr^2}\, ds, \tag{104}$$

where the second line substitutes $s = t + u$ in the inner integral and the third exchanges the order of integration over $0 < t < s < \infty$. The bracket is the convolution $\int_0^s (\pi^{-1/2} t^{-1/2} e^{-\kappa^2/(4t)}) \cdot 1 \, dt$ of the Yukawa weight with the constant weight of $1/r^2$. Multiplying by $\boldsymbol{r}\boldsymbol{r}^{\mathsf{T}}$ and using

$$K_\kappa(\boldsymbol{r}) = e^{-\kappa r} \frac{\boldsymbol{r}\boldsymbol{r}^{\mathsf{T}}}{r^3} \tag{105}$$

gives Eq. (2).

Closed form of the weight. Let $\varphi(s)$ denote the right-hand side of the second equality in Eq. (3). Differentiation gives

$$\frac{d}{ds}\left[\frac{2\sqrt{s}}{\sqrt{\pi}} e^{-\kappa^2/(4s)}\right] = \frac{1}{\sqrt{\pi}}\left(s^{-1/2} + \frac{\kappa^2}{2} s^{-3/2}\right) e^{-\kappa^2/(4s)}, \tag{106}$$

$$\frac{d}{ds}\left[\kappa\left[\operatorname{erf}\left(\frac{\kappa}{2\sqrt{s}}\right) - 1\right]\right] = -\frac{\kappa^2}{2\sqrt{\pi}} s^{-3/2} e^{-\kappa^2/(4s)}, \tag{107}$$

so $\varphi'(s) = \pi^{-1/2} s^{-1/2} e^{-\kappa^2/(4s)}$. Since $\varphi(s) \to 0$ as $s \to 0^+$, φ coincides with the integral in Eq. (3).

Properties of the weight. The integrand in Eq. (3) is positive and bounded by $\pi^{-1/2} t^{-1/2}$. Hence ρ_κ is positive and nondecreasing on $(0, \infty)$, and

$$\rho_\kappa(s) \le \frac{2\sqrt{s}}{\sqrt{\pi}}. \tag{108}$$

In particular, $\rho_\kappa(\xi^2) > 0$.

B Closed-form real-space coefficients

Let

$$z = \frac{\kappa}{2\xi}, \tag{109}$$

and define

$$c_{\kappa,\xi} = \rho_\kappa(\xi^2) = \frac{2\xi}{\sqrt{\pi}} e^{-z^2} + \kappa\left[\operatorname{erf}(z) - 1\right]. \tag{110}$$

The long-range scalar coefficient can be written as

$$\theta_L(r) = Y_\kappa^L(r) - c_{\kappa,\xi} e^{-\xi^2 r^2}, \tag{111}$$

where

$$Y_\kappa^L(r) = \frac{1}{2r}\left[e^{-\kappa r} \operatorname{erfc}(z - \xi r) - e^{\kappa r} \operatorname{erfc}(z + \xi r)\right]. \tag{112}$$

Similarly,

$$\theta_S(r) = Y_\kappa^S(r) + c_{\kappa,\xi} e^{-\xi^2 r^2}, \tag{113}$$

with

$$Y_\kappa^S(r) = \frac{1}{2r}\left[e^{-\kappa r} \operatorname{erfc}(\xi r - z) + e^{\kappa r} \operatorname{erfc}(\xi r + z)\right]. \tag{114}$$

Since

$$Y_\kappa^S(r) + Y_\kappa^L(r) = \frac{e^{-\kappa r}}{r}, \tag{115}$$

the identity $K_\kappa^S + K_\kappa^L = K_\kappa$ follows immediately.

C Localization estimates

This appendix proves Proposition 2.2 and Eq. (17).

Real-space decay. Fix $r > 0$. Since $r^2 e^{-sr^2} = -\frac{d}{ds} e^{-sr^2}$, integration by parts in

$$\theta_S(r) = r^2 \int_{\xi^2}^{\infty} \rho_\kappa(s) e^{-sr^2}\, ds \tag{116}$$

gives

$$\theta_S(r) = \rho_\kappa(\xi^2) e^{-\xi^2 r^2} + \int_{\xi^2}^{\infty} \rho_\kappa'(s) e^{-sr^2}\, ds. \tag{117}$$

The boundary term at infinity vanishes: by Eq. (108), $0 \le \rho_\kappa(s) e^{-sr^2} \le 2\pi^{-1/2}\sqrt{s}\, e^{-sr^2} \to 0$ as $s \to \infty$. By Appendix A,

$$\rho_\kappa'(s) = \frac{1}{\sqrt{\pi s}}\, e^{-\kappa^2/(4s)}, \tag{118}$$

so $0 \le \rho_\kappa'(s) \le (\sqrt{\pi}\,\xi)^{-1}$ for $s \ge \xi^2$. Inserting this bound in the last integral of Eq. (117) and using $\int_{\xi^2}^{\infty} e^{-sr^2}\, ds = e^{-\xi^2 r^2}/r^2$ gives

$$0 \le \theta_S(r) - \rho_\kappa(\xi^2) e^{-\xi^2 r^2} \le \frac{e^{-\xi^2 r^2}}{\sqrt{\pi}\,\xi r^2}. \tag{119}$$

Hence, for fixed $\kappa, \xi > 0$,

$$\theta_S(r) = e^{-\xi^2 r^2} \left[\rho_\kappa(\xi^2) + \mathcal{O}(r^{-2})\right], \qquad r \to \infty. \tag{120}$$

Since $\rho_\kappa(\xi^2) > 0$, this gives $\theta_S(r) \sim \rho_\kappa(\xi^2) e^{-\xi^2 r^2}$, and $K_\kappa^S(\boldsymbol{r}) = \theta_S(r) P(\boldsymbol{r})$ yields Eq. (10).

Fourier transform. For a Gaussian,

$$\mathcal{F}[e^{-sr^2}] = \left(\frac{\pi}{s}\right)^{3/2} e^{-k^2/(4s)}. \tag{121}$$

With the convention used here, multiplication by r_a corresponds to $i\,\partial/\partial k_a$ in Fourier space, so $\mathcal{F}[r_a r_b e^{-sr^2}] = -\partial_{k_a}\partial_{k_b}\mathcal{F}[e^{-sr^2}]$. Since $\partial_{k_a}\partial_{k_b} e^{-k^2/(4s)} = \big(k_a k_b/(4s^2) - \delta_{ab}/(2s)\big) e^{-k^2/(4s)}$,

$$\mathcal{F}[r_a r_b e^{-sr^2}] = \left(\frac{\pi}{s}\right)^{3/2} e^{-k^2/(4s)} \left[\frac{\delta_{ab}}{2s} - \frac{k_a k_b}{4s^2}\right]. \tag{122}$$

Integrating against $\rho_\kappa(s)$ over $(0, \xi^2)$, which is permitted because $\rho_\kappa(s) s^{-7/2}$ is integrable near $s = 0$, gives Eq. (11) with the stated $A_{\kappa,\xi}$ and $B_{\kappa,\xi}$. Differentiating $A_{\kappa,\xi}$ under the integral sign, $\partial_k e^{-k^2/(4s)} = -(k/(2s)) e^{-k^2/(4s)}$, gives

$$B_{\kappa,\xi}(k) = \frac{1}{k}\frac{d}{dk} A_{\kappa,\xi}(k), \qquad k > 0. \tag{123}$$

Endpoint asymptotics. Let $k > 0$ and, for $m \in \{5/2, 7/2\}$,

$$I_m(k) = \int_0^{\xi^2} \rho_\kappa(s) s^{-m} e^{-k^2/(4s)}\, ds, \tag{124}$$

so that $A_{\kappa,\xi}(k) = \frac{\pi^{3/2}}{2} I_{5/2}(k)$ and $B_{\kappa,\xi}(k) = -\frac{\pi^{3/2}}{4} I_{7/2}(k)$. Since

$$\frac{d}{ds} e^{-k^2/(4s)} = \frac{k^2}{4s^2} e^{-k^2/(4s)}, \tag{125}$$

we write $\rho_\kappa(s) s^{-m} = s^{-2}\, \rho_\kappa(s) s^{2-m}$ and integrate by parts, first on $[\varepsilon, \xi^2]$ and then letting $\varepsilon \to 0^+$:

$$I_m(k) = \frac{4\rho_\kappa(\xi^2)\xi^{4-2m}}{k^2} e^{-k^2/(4\xi^2)} - \frac{4}{k^2} \int_0^{\xi^2} \frac{d}{ds}\big[\rho_\kappa(s) s^{2-m}\big]\, e^{-k^2/(4s)}\, ds. \tag{126}$$

The boundary term at $s = 0$ vanishes: by Eq. (108),

$$0 \le \rho_\kappa(s) s^{2-m} e^{-k^2/(4s)} \le \frac{2}{\sqrt{\pi}}\, s^{5/2-m} e^{-k^2/(4s)} \longrightarrow 0, \qquad s \to 0^+, \tag{127}$$

because the exponential factor decays faster than any power of s.

To bound the remaining integral we use $\rho_\kappa'(s) \le \pi^{-1/2} s^{-1/2}$ and Eq. (108). For $0 < s \le \xi^2$,

$$\begin{aligned} \left| s^2 \frac{d}{ds}\big[\rho_\kappa(s) s^{2-m}\big] \right| &= \left| s^{4-m} \rho_\kappa'(s) + (2-m) s^{3-m} \rho_\kappa(s) \right| \\ &\le \frac{1 + 2|m-2|}{\sqrt{\pi}}\, s^{7/2-m} \le \frac{1 + 2|m-2|}{\sqrt{\pi}}\, \xi^{7-2m}, \end{aligned} \tag{128}$$

where the last step uses $7/2 - m \ge 0$. Hence, with $\int_0^{\xi^2} s^{-2} e^{-k^2/(4s)}\, ds = (4/k^2) e^{-k^2/(4\xi^2)}$,

$$\left| \frac{4}{k^2} \int_0^{\xi^2} \frac{d}{ds}\big[\rho_\kappa(s) s^{2-m}\big]\, e^{-k^2/(4s)}\, ds \right| \le \frac{16\,(1 + 2|m-2|)\,\xi^{7-2m}}{\sqrt{\pi}\, k^4} e^{-k^2/(4\xi^2)}, \tag{129}$$

which is $\mathcal{O}\big(k^{-4} e^{-k^2/(4\xi^2)}\big)$. The same bound shows that the integral in Eq. (126) converges absolutely, which justifies the limit $\varepsilon \to 0^+$. Since $\rho_\kappa(\xi^2) > 0$,

$$I_m(k) = \frac{4\rho_\kappa(\xi^2)\xi^{4-2m}}{k^2} e^{-k^2/(4\xi^2)} \left[1 + \mathcal{O}(k^{-2})\right], \qquad k \to \infty. \tag{130}$$

With $m = 5/2$ and $m = 7/2$,

$$A_{\kappa,\xi}(k) = \frac{2\pi^{3/2}}{\xi k^2} \rho_\kappa(\xi^2) e^{-k^2/(4\xi^2)} \left[1 + \mathcal{O}(k^{-2})\right], \tag{131}$$

$$k^2 B_{\kappa,\xi}(k) = -\frac{\pi^{3/2}}{\xi^3} \rho_\kappa(\xi^2) e^{-k^2/(4\xi^2)} \left[1 + \mathcal{O}(k^{-2})\right]. \tag{132}$$

Thus

$$\frac{A_{\kappa,\xi}(k)}{k^2 B_{\kappa,\xi}(k)} = -\frac{2\xi^2}{k^2} \left[1 + \mathcal{O}(k^{-2})\right], \qquad \lambda_\parallel(k) = A_{\kappa,\xi}(k) + k^2 B_{\kappa,\xi}(k) = k^2 B_{\kappa,\xi}(k) \left[1 + \mathcal{O}(k^{-2})\right], \tag{133}$$

which is Eq. (14); in particular $\lambda_\parallel(k) < 0$ for all sufficiently large k. The eigenvalues of the symmetric matrix $\widehat{K}_\kappa^L(\boldsymbol{k})$ are $\lambda_\perp(k) = A_{\kappa,\xi}(k) > 0$ (twice) and $\lambda_\parallel(k)$, so its spectral norm is $\max\{\lambda_\perp(k), |\lambda_\parallel(k)|\}$. Because $\lambda_\perp(k)/|\lambda_\parallel(k)| = \mathcal{O}(k^{-2})$, the norm equals $|\lambda_\parallel(k)|$ for large k, which gives Eq. (15).

Tail quantities. Since $\|\theta_S(r)P(\boldsymbol{r})\|_2 = \theta_S(r)$, spherical coordinates give

$$E_S(r_c) = 4\pi \int_{r_c}^{\infty} r^2 \theta_S(r)\, dr, \qquad E_F(k_c) = \frac{1}{2\pi^2} \int_{k_c}^{\infty} k^2 \left\| \widehat{K}_\kappa^L(k) \right\|_2 dk. \tag{134}$$

For $c > 0$, integration by parts gives

$$\int_{x_0}^{\infty} x^2 e^{-cx^2}\, dx = \frac{x_0}{2c} e^{-cx_0^2} + \frac{1}{2c} \int_{x_0}^{\infty} e^{-cx^2}\, dx, \tag{135}$$

and the last integral is at most $e^{-cx_0^2}/(2cx_0)$. Hence

$$\int_{r_c}^{\infty} r^2 e^{-\xi^2 r^2}\, dr \sim \frac{r_c}{2\xi^2} e^{-\xi^2 r_c^2}, \qquad \int_{k_c}^{\infty} k^2 e^{-k^2/(4\xi^2)}\, dk \sim 2\xi^2 k_c e^{-k_c^2/(4\xi^2)}. \tag{136}$$

Because the integrands are asymptotically equivalent to $\rho_\kappa(\xi^2) e^{-\xi^2 r^2}$ and $(\pi^{3/2}/\xi^3)\rho_\kappa(\xi^2) e^{-k^2/(4\xi^2)}$, respectively, the tails inherit these rates:

$$E_S(r_c) \sim 4\pi \rho_\kappa(\xi^2) \frac{r_c}{2\xi^2} e^{-\xi^2 r_c^2}, \qquad E_F(k_c) \sim \frac{\rho_\kappa(\xi^2)}{2\sqrt{\pi}\,\xi^3}\, 2\xi^2 k_c e^{-k_c^2/(4\xi^2)}, \tag{137}$$

which is Eq. (17).

D Pair-response operator bounds

Lemma D.1 (Pair-response perturbation)**.** *Let Γ and Γ' be built by the degree-minus-weight construction, $(\Gamma X)_i = \sum_{j\neq i} K_{ij}(x_i - x_j)$, from pair responses K_{ij} and K'_{ij}, and let $E_{ij} = K'_{ij} - K_{ij}$. If $E_{ij} = E_{ji} = E_{ij}^{\mathsf{T}}$ for all $i \neq j$, then*

$$\|\Gamma' - \Gamma\|_2 \leq 2 \max_i \sum_{j\neq i} \|E_{ij}\|_2. \tag{138}$$

Proof. The difference $\Gamma' - \Gamma$ is symmetric, and pairing the (i,j) and (j,i) terms gives

$$X^{\mathsf{T}}(\Gamma' - \Gamma)X = \frac{1}{2} \sum_{i\neq j} (x_i - x_j)^{\mathsf{T}} E_{ij} (x_i - x_j). \tag{139}$$

Using $|x_i - x_j|^2 \leq 2|x_i|^2 + 2|x_j|^2$ and the symmetry in i, j,

$$\left| X^{\mathsf{T}}(\Gamma' - \Gamma)X \right| \leq 2 \sum_i |x_i|^2 \sum_{j\neq i} \|E_{ij}\|_2 \leq 2 \max_i \sum_{j\neq i} \|E_{ij}\|_2\, |X|^2. \tag{140}$$

For a symmetric operator the spectral norm is the supremum of the quadratic form over unit vectors. □

The hypothesis holds for every even, pointwise symmetric pair kernel. It also holds for the particle–mesh response, because an even, real-symmetric influence function makes each block of W_h real and symmetric and $(W_h)_{ij} = (W_h)_{ji}$. Lemma D.1 with $E_{ij} = R_{k_c}(q_i - q_j)$ gives Eq. (25). Since $|e^{i\boldsymbol{k}\cdot\boldsymbol{r}}| = 1$, every term satisfies $\|R_{k_c}(\boldsymbol{r})\|_2 \leq V^{-1} \sum_{|\boldsymbol{k}|>k_c} \|\widehat{K}_\kappa^L(\boldsymbol{k})\|_2$, and the sum over $j \neq i$ has $N - 1$ terms, which gives Eq. (26). Lemma D.1 with $E_{ij} = K_{ij}^h - K_{ij}$ gives Eq. (38).

Proof of Proposition 3.2. Both terms in Eq. (40) are symmetric. Since $\widetilde{\Gamma}_h$ has zero block-row sums, it annihilates $\mathcal{T}$, and by symmetry it maps $\operatorname{Range}\Pi = \mathcal{T}^\perp$ into itself. Hence $\Gamma_h t = \widetilde{\Gamma}_h t + \delta_h \Pi t = 0$ for $t \in \mathcal{T}$. For arbitrary X, write $Y = \Pi X$. Then

$$X^\mathsf{T}\Gamma_h X = Y^\mathsf{T}\widetilde{\Gamma}_h Y + \delta_h |Y|^2 \geq \left(\lambda_{\min}\big(\widetilde{\Gamma}_h|_{\operatorname{Range}\Pi}\big) + \delta_h\right)|Y|^2 \geq 0 \tag{141}$$

by the choice of δ_h. Finally, $\|\Gamma_h - \Gamma\|_2 \leq \|\widetilde{\Gamma}_h - \Gamma\|_2 + \delta_h\|\Pi\|_2 \leq \epsilon_{\mathrm{op}} + \delta_h$ by Eq. (38) and $\|\Pi\|_2 = 1$. □

E Particle–mesh alias response

Single-mode pair response. Spreading a particle at q onto the mesh and taking the discrete Fourier transform produces, for a retained mode $\boldsymbol{k}$, the deposited amplitude $S_h(\boldsymbol{k}, q)$: sampling the window on the mesh folds all aliases $\boldsymbol{k}_n$ onto $\boldsymbol{k}$. Because the assignment window is real and even, $\widehat{W}_h$ is real, and gathering the field of mode $\boldsymbol{k}$ at q_i produces the factor $\overline{S_h(\boldsymbol{k}, q_i)}$. Hence the contribution of mode $\boldsymbol{k}$ to $(W_h)_{ij}$ is proportional to

$$\overline{S_h(\boldsymbol{k}, q_i)}\, G_h(\boldsymbol{k})\, S_h(\boldsymbol{k}, q_j) = H_h(\boldsymbol{k})\, e^{i\boldsymbol{k}\cdot(q_i - q_j)}\,(1 + \alpha_i^*)\,(1 + \alpha_j)\,, \tag{142}$$

with the same normalization as the exact contribution $\widehat{K}_\kappa^L(\boldsymbol{k})e^{i\boldsymbol{k}\cdot(q_i-q_j)}$. Subtracting the exact contribution and expanding the product gives Eq. (51).

B-spline alias ratio. Write $x_\ell = k_\ell h/2$. The ℓ-th component of $\boldsymbol{k}_n$ satisfies $(\boldsymbol{k}_n)_\ell h/2 = x_\ell + \pi n_\ell$, and $\sin(x_\ell + \pi n_\ell) = (-1)^{n_\ell}\sin x_\ell$. Therefore

$$\left|\frac{\widehat{W}_h(\boldsymbol{k}_n)}{\widehat{W}_h(\boldsymbol{k})}\right| = \prod_{\ell : n_\ell \neq 0} \left|\frac{k_\ell h/2}{k_\ell h/2 + \pi n_\ell}\right|^p. \tag{143}$$

For a mode resolved by the mesh, $|x_\ell| \leq \pi/2$, so $|x_\ell + \pi n_\ell| \geq \pi|n_\ell|/2$ and each factor is at most $(h|k_\ell|/\pi)^p |n_\ell|^{-p}$, where $h|k_\ell|/\pi \leq 1$ and $|k_\ell| \leq |\boldsymbol{k}|$. Every $n \neq 0$ has at least one nonzero component, so the product of the factors $(h|k_\ell|/\pi)^p$ is at most $(h|\boldsymbol{k}|/\pi)^p$ and

$$|\alpha_h(\boldsymbol{k}, q)| \leq \left(\frac{h|\boldsymbol{k}|}{\pi}\right)^p \sum_{n \neq 0} \prod_{\ell : n_\ell \neq 0} |n_\ell|^{-p} = \left[\left(1 + 2\sum_{m=1}^{\infty} m^{-p}\right)^3 - 1\right]\left(\frac{h|\boldsymbol{k}|}{\pi}\right)^p, \tag{144}$$

uniformly in q. The series converges for $p \geq 2$, which gives $\alpha_h(\boldsymbol{k}, q) = \mathcal{O}((h|\boldsymbol{k}|)^p)$.

Mesh law. For ideal deconvolution, $D_h = 0$, and the bound on α_h gives, for each pair,

$$\|E_{ij}^h\|_2 \lesssim \frac{1}{V}\sum_{|\boldsymbol{k}| \leq k_c} \|\widehat{K}_\kappa^L(\boldsymbol{k})\|_2 \left[(h|\boldsymbol{k}|)^p + (h|\boldsymbol{k}|)^{2p}\right], \tag{145}$$

with a constant depending only on p. Summing over $j \neq i$ multiplies the bound by $N - 1$. The lattice sum $V^{-1}\sum_{\boldsymbol{k}} |\boldsymbol{k}|^p \|\widehat{K}_\kappa^L(\boldsymbol{k})\|_2$ is a Riemann sum for $M_p(\kappa, \xi)$ and is finite by the Gaussian decay in Eq. (15). Since $h|\boldsymbol{k}| \leq \sqrt{3}\,\pi$ on the resolved modes, the h^{2p} term is subdominant, which gives Eq. (58) for a fixed configuration and cutoff.

F Finite-time thermostat

Exact covariance. For frozen positions, consider the Ornstein–Uhlenbeck dynamics

$$dp = -\frac{1}{m}\Gamma_h p\, dt + \left(2\beta^{-1}\Gamma_h\right)^{1/2} dW_t, \tag{146}$$

where W_t is a standard Wiener process. Over a step of length Δt, the solution is $p(\Delta t) = R_{\Delta t}(\Gamma_h)p(0) + \eta$ with a Gaussian increment η of mean zero. By the Itô isometry and the spectral calculus,

$$\mathrm{Cov}(\eta) = \int_0^{\Delta t} e^{-2\tau\Gamma_h/m}\, 2\beta^{-1}\Gamma_h\, d\tau = \beta^{-1}m\left(I - R_{\Delta t}(\Gamma_h)^2\right) = S_{\Delta t}(\Gamma_h)^2, \tag{147}$$

where both sides vanish on the null space of Γ_h. Since $f_{\Delta t}(\lambda)^2 = \beta^{-1}m(1 - e^{-2\Delta t\lambda/m})$, we have $S_{\Delta t}(\Gamma_h) = f_{\Delta t}(\Gamma_h)$, and the update $p \mapsto R_{\Delta t}(\Gamma_h)p + f_{\Delta t}(\Gamma_h)\xi$ reproduces the exact frozen-configuration transition law.

Momentum conservation. Let $t \in \mathcal{T}$. Since $\Gamma_h t = 0$, $R_{\Delta t}(\Gamma_h)t = t$ and $f_{\Delta t}(\Gamma_h)t = f_{\Delta t}(0)t = 0$. Both operators are symmetric, so

$$t^{\mathsf{T}}p^{+} = \left(R_{\Delta t}(\Gamma_h)t\right)^{\mathsf{T}} p + \left(f_{\Delta t}(\Gamma_h)t\right)^{\mathsf{T}} \xi = t^{\mathsf{T}}p \tag{148}$$

for every noise sample. The Lanczos approximation retains this property because the starting vector $z = \Pi\xi$ and the reconstructed increment are projected onto $\mathrm{Range}\,\Pi = \mathcal{T}^{\perp}$, and the center-of-mass component is copied exactly through the damping step.

Invariance of the conditional Maxwell law. Suppose p is Gaussian with fixed total momentum and covariance $\beta^{-1}m\Pi$. Since $R_{\Delta t}(\Gamma_h)$ commutes with Π and equals the identity on $\mathcal{T}$, $I - R_{\Delta t}(\Gamma_h)^2 = \Pi - R_{\Delta t}(\Gamma_h)^2\Pi$. Hence

$$\mathrm{Cov}(p^{+}) = \beta^{-1}m\left(R_{\Delta t}(\Gamma_h)^2\Pi + \Pi - R_{\Delta t}(\Gamma_h)^2\Pi\right) = \beta^{-1}m\Pi, \tag{149}$$

and the total momentum is unchanged. The conditional Maxwell law is therefore invariant under the thermostat step.

Range-start function. Expanding the exponential gives $f_{\Delta t}(\lambda)^2 = 2\beta^{-1}\Delta t\,\lambda + \mathcal{O}(\lambda^2)$ as $\lambda \to 0^{+}$. Hence the constant in the range-start analysis is $C = (2\beta^{-1}\Delta t)^{1/2}$, and $g(\lambda) = f_{\Delta t}(\lambda)/\lambda$ has the stated $\lambda^{-1/2}$ singularity.

G Technical details of the operator experiment

Configurations. Cases A and B are single draws of independent uniform positions (seeds 7 and 13) with no minimum distance. The perturbed lattices displace every site of a bcc lattice of 4^3 cells ($N = 128$; fcc for $N = 256$) by an isotropic Gaussian of standard deviation 0.1 times the nearest-neighbour distance and add a random global shift. The clustered configurations draw $N/16$ centres uniformly and 16 particles per centre from an isotropic Gaussian of width 0.5, rejecting any draw closer than 0.25 to a particle already placed; this avoids near-coincident pairs, whose $1/r$ kernel would make $\|\Gamma\|_2$ arbitrarily large. All generators are seeded, and the seeds are stored with the positions.

Wave-number rounding and the numerical floor. An earlier version of the coefficient routine evaluated $A_{\kappa,\xi}(k)$ and $B_{\kappa,\xi}(k)$ at $|\boldsymbol{k}|$ rounded to twelve decimals, which perturbs the input by up to 5×10^{-13} and produced an artificial floor near 10^{-13} in every Fourier-side comparison. The routine now uses the unrounded $|\boldsymbol{k}|$, and all Fourier-side quantities of Section 6, including the production operator, were recomputed; the reference, the configurations and the quadrature checks do not involve the routine. Table 4 compares the mesh-free Fourier sum ($k_c = 13\xi$) with the real-space image sum of θ_L on case A, one perturbed lattice and one uniform configuration at $\xi = 0.7$ and 1. The remaining floor, at most 7.6×10^{-16}, has the size of the float64 round-off of the Fourier sum and of the numerical uncertainty of the image sum; these two contributions were not separated further. It lies far below the fit floor 1.1×10^{-13} and every production error ($\geq 2.3 \times 10^{-10}$).

variant	range	source probed
$\|\boldsymbol{k}\|$ rounded to 12 decimals (earlier)	2.3×10^{-14}–1.5×10^{-13}	wave-number rounding
unrounded $\|\boldsymbol{k}\|$, float64 (current)	6.3×10^{-17}–7.6×10^{-16}	baseline
k from the integer shell $\|\boldsymbol{m}\|^2$	5.4×10^{-17}–8.5×10^{-16}	wave-number input
refined quadrature of $A_{\kappa,\xi}$, $B_{\kappa,\xi}$	6.5×10^{-17}–7.9×10^{-16}	quadrature
sum evaluated in 80-bit arithmetic	3.5×10^{-17}–4.0×10^{-16}	floating point
float64 minus 80-bit sum	4.2×10^{-17}–6.7×10^{-16}	floating point
image sum, $R = 40$ against $R = 45$	4.1×10^{-17}–4.3×10^{-16}	reference

Table 4: Distance between the mesh-free Fourier sum and the real-space image sum of the long-range part, relative to $\|\Gamma\|_2$. Source: `fourier_precision.csv`.

Cutoff sweeps. The real-space error is computed against the real-space lattice sum of θ_S over $|\boldsymbol{r} + \boldsymbol{m}L| \leq 45$, and the Fourier error against the real-space lattice sum of θ_L over the same images. The fits use $x \geq 1$ and points above the fit floor, and exclude steps at which a larger cutoff admits no new pair image or lattice shell (on the perturbed lattices for $\xi r_c \leq 1.25$, and for case A for $k_c/(2\xi) \leq 1$); seven such points fell inside a fit window. Below the fit floor the Fourier error levels off near $x = 6$ at the floor of Table 4; the real-space error, which compares two real-space sums over the same images, keeps falling in 25 of 28 curves.

Mesh scans. At $s = 4.1$ no mesh falls below the fit floor (smallest error 4.5×10^{-13}). At $s = 5.5$ the finest mesh for $p = 8$ does in two configurations, and the fit then takes the next four meshes. The three $N = 256$ checks ($p = 5, 7$, six meshes, four shifts) give slopes 5.12–5.39 and 7.44–7.58, inside the ranges of the main set.